MIL-A-23865A(SHIPS)
11 August 1964
SUPERSEDING
MIL-A-23865(SHIPS)
1 October 1963
(see 6.3)

MILITARY SPECIFICATION

ANTENNA GROUP AN/SRA-42( )

1. SCOPE

1.1 This specification establishes the minimum acceptable physical and performance requirements for
the design, manufacture, and testing of Antenna Group AN/SRA-42( ) It is intended that the antenna group be capable of remote tuning from the receiver location and of performing the functions of selecting and coupling LORAN signals at the LORAN C carrier frequency of 100 kilocycles (kc.) and in the standard LORAN frequency range between 1700 kc. and 2350 kc.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-901 - Shock Tests, H. I. (High-Impact); Shipboard Machinery Equipment and Systems, Requirements for.
MIL-E-16400 - Electronic Equipment, Naval Ship and Shore: General Specification.
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.
MIL-F-19207 - Fuseholders, Extractor Post Type, Blown Fuse Indicating and Nonindicating.
MIL-R-22732 - Reliability Requirements for Shipboard and Ground Electronic Equipment.
MIL-M-23313 - Maintainability Requirements for Shipboard and Shore Electronic Equipment and Systems.

STANDARDS

MILITARY

MIL-STD-108 - Definitions of and Basic Enclosures for Electric and Electronic Equipment.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, New York.)

3. REQUIREMENTS

3.1 Preproduction sample.- Prior to production, a preproduction sample should be tested as specified in 4.2 (see 6.2).

3.2 General requirements.- The equipment shall be in accordance with all provisions of MIL-E-16400, except as otherwise specified herein.

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3.2.1 Temperature.- Individual units of the antenna group shall be designed to meet the following temperature classifications:

Class 2 - Antenna unit and radio frequency tuner unit.
Class 4 - Antenna control unit.

3.2.2 Enclosure.- The antenna unit and the radio frequency tuner unit shall be watertight in accordance with MIL-STD-108. The antenna control unit shall be dripproof in accordance with MIL-STD-108.

3.2.3 Electrical design.- The nominal supply voltage and frequency shall be 115 volts and 60 cycles per second (c.p.s.), single phase.

3.2.3.1 Overload protection in the primary power circuit shall be provided by fuses. Glow lamp type blown fuse indicators shall be provided. Fuseholders shall be the extractor post type as defined in MIL-F-19207.

3.2.3.2 External connectors shall be provided in accordance with MIL-E-16400 for all electrical and electronic inter-unit connections.

3.2.4 Size and weight.- The volume of the radio frequency tuner unit shall be not greater than 1000 cubic inches; the total weight shall not exceed 40 pounds. The volume of the antenna control unit shall be not greater than 80 cubic inches; the weight shall not exceed 5 pounds.

3.3 Reliability.- The reliability level for this antenna group shall be 1000 hours, mean-time-between-failures (MTBF). For test purposes, demonstration of attainment of a reliability level of 300 hours MTBF with a 90-percent confidence level is required.

3.4 Maintainability.- A maintainability program in accordance with MIL-M-23313 shall be established. The equipment repair time shall not exceed 10.0 minutes.

3.5 Controls, indicators, and panel layout.- Tactile identification and control locking devices are not required.

3.6 Elapsed time meters.- Elapsed time meters are not required.

3.7 Shock.- The antenna group shall conform to the grade A, class 1, lightweight shock requirements of MIL-S-901. The Type A test shall be conducted to show shock resistivity.

3.8 Design requirements.-

3.8.1 Design objective.- The basic design objective of the antenna group is to provide a physically small capacitive probe (antenna) which is tunable at the desired LORAN frequency. The induced voltage on the ship's mast or superstructure is sampled by this tuned probe and fed to the receiver. It is required that the tuner be housed at the base of the probe and that the tuning be accomplished remotely from the receiver location.

3.8.2 Antenna unit.- The antenna unit shall consist of a corrosion-resistant, tubular, whip-type antenna with ceramic bowl insulator and associated mounting hardware. The whip antenna shall have an outside diameter no greater than 1 inch, and shall have a top cap, or other suitable provision to prevent water entrance into the whip. Tapering is permitted. The unit shall be mounted on the enclosure for the radio frequency tuner unit.

3.8.3 Radio frequency tuner unit.- The radio frequency tuner unit shall consist of a watertight enclosure housing an air-cored inductor, a tuning capacitor, drive motor, and associated parts. All parts for tuning the whip antenna shall be mounted on a separate pull-out chassis for ease of maintenance. Installation mounting of the tuner unit enclosure shall be independent of the pull-out chassis. Provisions shall be made to allow mounting with the whip in the vertical or horizontal positions.

3.8.3.1 Tuner details.- The radio frequency tuner shall include an antenna inductor, a tuning capacitor, a capacitor drive motor, and a neon tube voltage limiter. The antenna shall be connected to one end of the antenna inductor and the other end of the inductor shall be grounded. The tuning capacitor shall be placed across the antenna inductor. The voltage limiter, consisting of two NE51 or suitable standard type neon glow lamps in series, shall be placed across the antenna inductor. The action of the voltage limiter shall be

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to limit voltages across the antenna inductor in excess of 140 volts. The output of the tuner shall be connected to a 50-ohm transmission line by a separate inductor or by tapping the antenna inductor. The proper bandwidth shall be obtained by overcoupling to the receiver (nominal 50-ohms impedance). No resistive elements are allowed.

3.8.3.1.1 Capacitor drive motor.- An electric motor, capable of clockwise and counterclockwise rotation shall provide the capacitor drive. Shaft speed of the capacitor drive motor shall be 1 revolution per minute (r.p.m.) when used with a 60-c.p.s. power source. Backlash shall be minimized. Torque shall not be available to the capacitor drive immediately after primary power to the motor is removed. The torque available from the motor shaft shall exceed the torque necessary to turn the variable capacitor by a factor of four or more.

3.8.3.2 100-kc. tuning.- A fixed network for tuning the antenna group at 100 kc. shall be included in the radio frequency tuner unit. Means for switching the frequency selection between the 100-kc. channel and the standard LORAN frequency range of 1700 kc. and 2350 kc. shall be provided.

3.8.4 Antenna control unit.- The antenna control unit shall consist of an enclosure housing a motor drive control. Two push-button sensitive type switches shall be used to control the power to the reversible tuner motor. A switch control, mounted near the push-button switches, shall enable the LORAN operator to tune the antenna group at the fixed frequency of 100 kc. or in the frequency range between 1700 kc. and 2350 kc. Fuses for the input power line shall be mounted on the antenna control unit enclosure, and shall be accessible from the outside.

3.8.5 Radio interference and undesired radiation.-

3.8.5.1 Provisions shall be made to minimize r.f. pulses caused by sparking at the motor control switches.

3.8.5.2 Noise.- Noise shall not be detectable in the LORAN receiver due to operation of the motor when the antenna tuning is being accomplished.

3.8.6 Sensitivity.- The voltage drop ratio, obtained in accordance with 4.6.1 shall not exceed 3.3.

3.8.7 Bandwidth.- The bandwidth at the carrier frequencies of 1750, 1850, 1900, and 1950 kc. shall be no less than 55 kc. and no greater than 70 kc. measured at the 3-db points.

3.8.7.1 The bandwidth at the carrier frequency of 100 kc. shall be no less than 25 kc. and no greater than 30 kc. measured at the 3-db points.

3.8.8 Compatibility.- The antenna group shall be designed to operate with LORAN receiving equipment such as the AN/UPN-12( ) and AN/UPN-15( ) LORAN Receiving Sets.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction inspection.- Preproduction inspection shall consist of the preproduction inspection of MIL-E-16400, except the accelerated life and water cooling tests, and the tests specified in 4.6.

4.2.1 The contractor shall submit an outline of tests and test procedures to the bureau or agency concerned for approval prior to setting up the test equipment for test. After preproduction inspection, a report of the test results shall be forwarded to the bureau or agency concerned for review and approval in accordance with MIL-E-16400.

4.3 Production inspection.- Production inspection shall be conducted as specified in MIL-E-16400. Production inspection shall also include the tests specified in 4.6.

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4.4 Production control inspection.- Production control inspection shall be made on each sample selected in accordance with 4.4.1. Production control inspection shall be combined with the environmental tests and shall consist of the following tests of MIL-E-16400:

(a) Weight and dimensions.
(b) Supply line voltage and frequency.
(c) Heat.
(d) Power.
(e) Controls and control circuits.
(f) Enclosure.
(g) Salt spray.
(h) Shock.

4.4.1 Sampling for production control inspection.- Sample units shall be selected at random for production control inspection as follows:

(a) First ten production units - One unit.
(b) Following production units to the end of the contract - One unit out of every 20 or fraction thereof.

4.5 The tests of 4.6 shall be utilized to determine the capability of the equipment in those cases where the quality assurance provisions require measurement of performance parameters.

4.6 Test procedures.-

4.6.1 Sensitivity test.- Voltage drop shall be determined as follows:

(a) Connect a 50-ohm non-inductive resistor across the tuner output terminals.
(b) Connect one terminal of a 30-$\mu\mu$f capacitor to the antenna terminal of the antenna inductor.
(c) Terminate a signal generator with a 1-ohm non-inductive resistor.
(d) Connect the other terminal of the 30-$\mu\mu$f capacitor to the ungrounded end of the 7-ohm resistor.
(e) Place a 5000-ohm resistor in series with the input to a transmission line connection to a receiver (at the opposite end from the receiver).
(f) Ground the outer conductor of the transmission line to the tuner chassis.
(g) Connect the other end of the 5000-ohm resistor to the ungrounded end of the 50-ohm load resistor.
(h) Tune signal generator, tuner and receiver to 1750 kc., set a level on the receiver and record the signal generator output.
(i) Disconnect the 5000-ohm resistor from the 50-ohm resistor and connect it to the function of the 30-$\mu\mu$f capacitor and 1-ohm resistor.
(j) Reset signal generator level to make the receiver level the same as it was in (h).
(k) Take the signal generator level noted in (h) and divide it by the level noted in (j). The ratio of level (h) divided by level (j) is the voltage drop.

4.6.1.1 The sensitivity test shall be repeated with the signal generator, tuner, and receiver set at 1950 kc. and 100 kc.

4.6.1.2 A high impedance r.f. voltmeter may be used in lieu of the receiver.

4.6.2 Bandwidth test.- Bandwidth shall be measured as follows:

(a) Connect a 50-ohm non-inductive resistor across the tuner output terminals.
(b) Connect one terminal of a 30-$\mu\mu$f capacitor to the antenna terminal of the antenna inductor.
(c) Leaving tuner fixed, tune signal generator and receiver above and below 1750 kc. while maintaining the same receiver level by increasing the signal generator output.
(d) Record the frequencies above and below 1750 kc. at which the signal generator output is 1.41 times the output at 1750 kc. The difference of the two frequencies is the bandwidth.

4.6.2.1 The bandwidth test shall be repeated at 1950 kc. and 100 kc.

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4.6.3 Operating test.- An operating test shall be conducted as follows:

(a) Connect a signal generator to the antenna terminal of the radio frequency tuner unit through a 30-$\mu\mu$f capacitor.
(b) Place a 50-ohm non-inductive resistor across the output terminals and connect a receiver to the ungrounded terminal of the 50-ohm resistor (without the 5000-ohm decoupling resistor).
(c) Tune signal generator and receiver to 1700 kc. and tune the radio frequency tuner by use of the antenna control unit.
(d) Ascertain that the tuning capacitor is not completely meshed.
(e) Tune signal generator and receiver to 2350 kc. Tune the radio frequency tuner by use of the antenna control unit.
(f) Ascertain that the tuning capacitor is not completely unmeshed.

4.6.4 Noise test.- The noise test shall be conducted as follows:

(a) Set up as in 4.6.3 with the exception that the signal generator is removed.
(b) Ground the end of the 30-$\mu\mu$f capacitor which went to the signal generator.
(c) Set a reference noise level on the receiver.
(d) Tune the motor at least (revolution) with the receiver set at 1850 kc. No increase in receiver noise should be detected when the motor is energized.

4.6.5 Reliability test.- Reliability testing shall be conducted at environmental test level number 1. The noise and operating tests of paragraph 4.3 shall be conducted during the reference test period. The "on-off" cycling and input voltage test shall be conducted as specified in MIL-R-22732 with the exception that the following requirement is added: During the period that the equipment is energized, means to continuously operate the variable capacitor drive motor shall be provided. The direction of the motor shall be changed at a rate to allow complete coverage of the specified tunable frequency band during each clockwise and counterclockwise sweep of the motor.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts.-

5.1.1 Antenna group AN/SRA-42( ).-

5.1.1.1 Preservation and packaging.- Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation and may conform to the supplier's commercial practice when such meets these requirements.

5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical and mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice when such meets these requirements.

5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, contract or order number, contractor's name and destination.

5.1.2 Onboard repair parts.- Onboard repair parts shall be preserved and packaged level A; packed level C and marked in accordance with MIL-E-17555.

5.2 Domestic shipment and storage or overseas shipment.- The requirements and levels of preservation and packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels for protection during domestic shipment and storage or overseas shipment which may be required when procurement is made.

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(5.2.1.1 Preservation, packaging, packing and marking.- The equipment and accessories, repair parts, and technical publications shall be preserved and packaged level A or C; packed level A or B as specified (see 6.1) and marked in accordance with MIL-E-17555.)

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preservation and packaging, packing, and marking requirements if other than specified in 5.1 (see 5.2).

6.2 Preproduction.- Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.3 CHANGES FROM PREVIOUS ISSUE. - THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

Preparing activity:
Navy - Ships
(Project 5985-N181Sh)

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4063, WHICH IS OBSOLETE

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